

# Online Learning Applications

## Project Requirements

# Recap on evaluation

The evaluation is composed of two parts:

## Written exam

- Scheduled on the official calls (see Online Services)
- max 16 points
- Questions about theory

## Project

- Three scheduled project presentations (see last slides)
- max 16 points
- Groups of at most 4-5 students

# Recap on evaluation

The evaluation is composed of two parts:

## Written exam

- Scheduled on the official calls (see Online Services)
- max 16 points
- Questions about theory

## Project

- Three scheduled project presentations (see last slides)
- max 16 points
- Groups of at most 4-5 students **⚠ If you can't find a group, send me an email by Friday (17/04) and I will try to match you with other students without a group.**
- The goal is to develop algorithms for a complex problem
- Includes modeling and coding

## Overview of the project

# Goal

The goal of the project is to design online learning algorithms to bid on **multiple advertising campaigns** under **budget constraints**.

# Setting

An agency is running multiple advertisement campaigns for different products. Its goal is to choose bids dynamically.

## Parameters

- Number of rounds  $T$
- Number of campaigns  $N$
- For each  $i \in \{1, \dots, N\}$ , value  $v_i$  of winning the slot in the  $i$ -th campaign
- Set of possible bids  $\mathcal{B}$  (**small and discrete set**)
- Budget  $B$

## Auction model

- For each campaign, at each round a different **first-price auction** is run to win the only available slot
- The highest competing bid  $m_i$  (**different for each campaign!**) fully characterizes the competing bids

# Interaction

At each round  $t \in T$ :

- 1 The advertiser sets a bid  $b_i$  for each campaign  $i \in \{1, \dots, N\}$
- 2 The other bidders select a competing bid  $m_i$  for each campaign  $i \in \{1, \dots, N\}$
- 3 The advertiser wins all the auctions such that  $b_i \geq m_i$
- 4 Observe **Feedback**: set of won auctions
- 5 For each won auction, the advertiser gets utility  $v_i - b_i$  and pays cost  $b_i$

## Additional constraint: non-compatible campaigns

The agency cannot simultaneously run campaigns for direct competitors' products (e.g., Coca-Cola and Pepsi).

### Parameters

- A conflict graph. In this graph, each node represents a campaign, and an edge between two nodes means those campaigns are mutually exclusive (non-compatible) and cannot be bid on (and hence displayed) in the same round  $t$ .

Requirement 1: Single campaign and stochastic environment

# Environment

Build a **stochastic** environment:

- A distribution over the highest competing bid

# Design algorithms for a single campaign and stochastic environment

## Algorithm

Build a bidding strategy using UCB1 **ignoring the budget constraint**.

## Algorithm

Build a bidding strategy extending UCB1 to handle the **budget constraint**.

## Requirement 2: Multiple campaigns and stochastic environment

# Environment

Build a **stochastic** environment:

- A joint distribution over the highest competing bids for each campaign

# Design algorithms for multiple campaigns and stochastic environment

## Algorithm

Build a bidding strategy using Combinatorial-UCB **with budget constraint**.

# Design algorithms for multiple campaigns and stochastic environment

## Algorithm

Build a bidding strategy using Combinatorial-UCB **with budget constraint**.

## Hint

Extend the “UCB-like” approach that we saw for a single campaign to multiple campaigns.

## Requirement 3: Best-of-both-worlds with multiple campaigns

# Environment

Use the **stochastic** environment already designed:

- A joint distribution over the highest competing bids for each campaign

Build a **highly non-stationary** environment. At a high level, it should include:

- A non-stochastic sequence of highest competing bids for each campaign (e.g., sampled from a distribution that changes **quickly** over time)

# Design best-of-both-worlds algorithms with multiple campaigns

## Algorithm

Build a bidding strategy using a primal-dual method **with budget constraint**.

# Design best-of-both-worlds algorithms with multiple campaigns

## Algorithm

Build a bidding strategy using a primal-dual method **with budget constraint**.

## Hint

Design a primal regret minimizer for the specific problem under study.

# Design best-of-both-worlds algorithms with multiple campaigns

## Algorithm

Build a bidding strategy using a primal-dual method **with budget constraint**.

## Hint

Design a primal regret minimizer for the specific problem under study.

 For this requirement, you can assume **full feedback**, i.e., to observe the highest competing bid for each auction.

Requirement 4: Slightly non-stationary environments with multiple campaigns

# Non-stationary environment

Build a **slightly non-stationary** environment for the pricing problem. At a high level:

- Rounds are partitioned in intervals
- In each interval the distribution of highest competing bids is fixed
- Each interval has a different distribution

## Algorithm

Extend the Combinatorial-UCB approach in Requirement 2 with

- **sliding window**
- **a change detector**

## Compare

Compare the performance of:

- Combinatorial-UCB with sliding window
- Combinatorial-UCB with the change detector
- The primal-dual method

## Deliverable and presentation

# Deliverable

Before the presentation you should deliver:

- Some slides describing the project and the results in detail:
  - ▷ **High level details** about the implemented algorithms
  - ▷ **graphs** of the empirical results
- A link to a GitHub repository with the code of the project

The presentation should be conducted as follows:

- You will have 20 minutes to deliver your presentation: additional 10 minutes will be used for questions and answers
- A detailed discussion of unexpected results is appreciated (e.g. the algorithm does not achieve a sublinear regret)

## Details about the project presentation schedule

You have three possible sessions to present the project in **July, September, December**;

- The deadline for the July project submission is **11th of July**
- The presentations will be in the days following the submission